Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 2026
Team ID	
Project Name	EV Market Intelligence Dashboard
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Language used to query, filter, transform, aggregate, and analyze data.	JavaScript (ES6), HTML5, CSS3	Single-page web dashboard; JavaScript handles filtering, aggregation and rendering with no backend needed.
2.	Database / Data Warehouse	Warehouse Used to store and manage structured business data for analysis and reporting.	In-memory JavaScript dataset (array of 40 EV models); CSV / Excel as source	Small illustrative dataset embedded in the page; can be swapped for a CSV export or database later.
3.	Data Ingestion	Used to load data from source files or external systems into the analytical environment.	Embedded data array in the page source (replaceable)	Simplest way to load data with no external connection; users replace the array with their own dataset.
4.	Data Transformation and Business Logic	Tools/Logic used to clean, join, filter, aggregate, data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	JavaScript functions for grouping, sums, weighted averages and growth calculation	Computes KPIs, market shares, sales-weighted range and rating, and 2021-2025 history from assumed growth rates.
5.	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	Chart.js 4.4 (doughnut, bar and line charts)	Lightweight interactive charts that update with filters and support light and dark themes.

6.	Front-End / User Interface	Interface through which users interact with dashboards, filters, parameters, and visual insights.	Responsive HTML/CSS page with persona tabs, filter dropdowns and a sortable table; hosted as a static web page	Easy for analysts to open and share as a link; no installation, works on desktop and mobile browsers.